

Derivation of the Standard Model Gauge Algebra, Per Generation Matter Content, and Hypercharge from the Hodge Complex on the 8×8 Torus

A Structural Result of the WIN Paradigm

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Abstract

We derive the Standard Model gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$, the per-generation matter content, and the Standard Model hypercharge assignment from the Hodge complex of differential forms on the 8×8 torus. The complex is $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with total dimension 256; the Hodge-Dirac operator $D = d + d^*$ satisfies $D^2 = \Delta$ exactly, without fermion doubling. At the self paired eigenvalue $\lambda = 4$, the content decomposes under D_4 as $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$. We prove: (i) the D_4 commutants on $\Omega^0|_{\lambda=4}$ and $\Omega^1|_{\lambda=4}$ have dimensions 27 and 100 with Lie algebra structures $\mathfrak{f}_3 \oplus \mathfrak{f}_1 \oplus \mathfrak{f}_2^2 \oplus \mathfrak{f}_3$ and $\mathfrak{f}_3^4 \oplus \mathfrak{f}_8$; (ii) the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ sits inside the commutant as a direct product subalgebra with closure residuals at machine precision; (iii) the vacuum zero state s is a color singlet and a weak doublet, with the color-singlet property forced by parity; (iv) the Hodge-Dirac evades Nielsen-Ninomiya, with zero modes equal to the Betti numbers of the torus and a $\pm$ -symmetric nonzero spectrum; (v) the 9 copies of the A_1 irrep split 3/3/3 across the three Hodge sectors, providing exactly three color triplets per generation; (vi) the assignment of these triplets to u_R^c, Q_L, d_R^c across the sectors gives the unique decomposition of the SM hypercharges as a linear combination of $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ with A_1 coefficients $(\alpha, \beta, \gamma) = (0, -1/6, -1/2)$ and B_1 coefficients $(0, 1/2, 1/2)$; (vii) all four anomaly cancellation conditions $\text{Tr}(Y) = \text{Tr}(Y^3) = \text{SU}(2)^2\text{U}(1) = \text{SU}(3)^2\text{U}(1) = 0$ vanish exactly. All numerical claims are verified by explicit algorithms included in the paper. The derivation is conditional on the sector coarse hypothesis (Section 9). Open problems: deriving the sector coarse hypothesis, three-generation structure, Yukawa couplings, and the V_{td} anomaly.

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1 Introduction

1.1 Motivation

The Standard Model (SM) of particle physics describes strong, weak, and electromagnetic interactions through the gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$ acting on chiral fermions. The gauge group, fermion content, and anomaly cancellation pattern are phenomenological inputs. No framework in the literature derives these from a discrete geometric substrate in a way the community accepts.

This paper presents a structural derivation within the WIN Paradigm, whose substrate is the Hodge complex of differential forms on the 8×8 periodic torus. The derivation establishes the gauge algebra, the vacuum structure, the per-generation matter content, the sector-triplet correspondence, and the SM hypercharges as theorems (conditional on the sector-coarse hypothesis).

1.2 Main results

- (i) Hodge–Dirac structure: $D = d + d^*$ with $D^2 = \Delta$ exactly (Theorem 2.2).
- (ii) $\lambda = 4$ content: $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$ (Theorem 3.1).
- (iii) Commutants: $\dim 27$ and $\dim 100$ (Theorem 4.2).
- (iv) Gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \Phi_1^5$ as an exact Lie subalgebra (Theorem 5.3).
- (v) Nielsen–Ninomiya evasion (Theorem 6.1).
- (vi) Vacuum s : color singlet, weak doublet (Theorems 7.2, 7.4).
- (vii) Sector-triplet correspondence: $3/3/3$ split (Theorem 8.1).
- (viii) Orientation- Z_2 theorem: SM hypercharges uniquely determined (Theorem 9.3).
- (ix) Anomaly cancellation: all four conditions vanish exactly (Theorem 10.2).
- (x) Non-equivariance: SM hypercharge is not D_4 -equivariant (Theorem 11.1).

1.3 Organization

Sections 2–3 establish the Hodge complex. Sections 4–5 derive the commutants and the gauge algebra. Section 6 proves NN evasion. Section 7 analyzes the vacuum. Sections 8–10 derive the sector-triplet correspondence, the orientation- Z_2 theorem, and verify anomaly cancellation. Section 11 proves non-equivariance. Sections 12–13 close. Appendix A collects all numerical algorithms.

2 The Hodge complex on the 8×8 torus

2.1 Substrate and differential forms

Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ be the periodic square lattice with $N = L^2 = 64$ sites and lattice spacing $a = 1$. The Hodge complex is

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \tag{1}$$

with total dimension $64 + 128 + 64 = 256$.

In local coordinates with lattice basis vectors e_1, e_2 , the exterior derivative d and codifferential d^* act as

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad (2)$$

$$(d_1 \omega)(x) = \omega_1(x) + \omega_2(x + e_1) - \omega_1(x + e_2) - \omega_2(x), \quad (3)$$

$$(d_0^* \omega)(x) = \sum_{\mu} [\omega_{\mu}(x) - \omega_{\mu}(x - e_{\mu})], \quad (4)$$

$$(d_1^* \chi)(x, \mu) = \pm [\chi(x) - \chi(x + e_{\mu})], \quad (5)$$

with signs chosen so that $d_0^* d_1^* = 0$. Nilpotency holds exactly: $d_1 d_0 = 0$.

Definition 2.1 (Hodge–Dirac). The Hodge–Dirac operator is $D = d + d^*$, acting on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$.

Theorem 2.2 (Hodge–Dirac square). *On the discrete torus,*

$$D^2|_{\Omega^k} = \Delta \cdot \mathbb{K}_{\Omega^k}, \quad k = 0, 1, 2, \quad (6)$$

where Δ is the scalar 5-point Laplacian

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x + e). \quad (7)$$

In particular, D^2 is block diagonal in the sector decomposition $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$.

Proof. Direct computation. The nilpotency $d_1 d_0 = 0$ forces $\{d, d^*\}$ to be block-diagonal, and on each form degree $\{d, d^*\} = \Delta$. Unlike the spinorial Dirac operator on the same lattice, there is no fermion doubling. $\square$

2.2 D_4 action

The dihedral group $D_4 = \langle r, s \mid r^4 = s^2 = (rs)^2 = e \rangle$ acts on Λ by the point group of the square:

$$r : (n_1, n_2) \mapsto (-n_2, n_1), \quad s : (n_1, n_2) \mapsto (n_1, -n_2), \quad (8)$$

modulo 8. On Ω^0 , D_4 permutes sites. On Ω^1 , D_4 acts with the vector twist:

$$r : (\omega_1, \omega_2) \mapsto (-\omega_2, \omega_1), \quad (9)$$

$$s : (\omega_1, \omega_2) \mapsto (\omega_1, -\omega_2). \quad (10)$$

Proposition 2.3. *The D_4 action preserves each sector and commutes with D^2 :*

$$[R, D^2] = [S, D^2] = 0, \quad [R, P_{\Omega^k}] = [S, P_{\Omega^k}] = 0, \quad (11)$$

where P_{Ω^k} is the projector onto Ω^k . Numerical verification: $\|[R, D^2]\| = \|[S, D^2]\| = 0$ exactly (Algorithm 1).

3 The $\lambda = 4$ sectors

The scalar Laplacian on Λ has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (12)$$

mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. At $\lambda = 4$, the scalar multiplicity is 14.

Theorem 3.1 ($\lambda = 4$ decompositions). *At the self-paired eigenvalue $\lambda = 4$, the Hodge complex decomposes under D_4 as*

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (14 \text{ states}), \quad (13)$$

$$\Omega^1|_{\lambda=4} = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E \quad (28 \text{ states}), \quad (14)$$

$$\Omega^2|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (14 \text{ states}), \quad (15)$$

giving a total of 56 states with D_4 content

$$9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (16)$$

Proof. The Ω^0 decomposition follows from the three D_4 orbits on the 14 modes:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\} : A_1 \oplus B_1, \quad (17)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\} : A_1 \oplus A_2 \oplus B_1 \oplus B_2 \oplus 2E, \quad (18)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\} : A_1 \oplus B_2 \oplus E. \quad (19)$$

For Ω^1 , the vector twist gives the tensor identity $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$. Using $A_i \otimes E = B_i \otimes E = E$ and $E \otimes E = A_1 \oplus A_2 \oplus B_1 \oplus B_2$, the Ω^1 content follows. The Ω^2 content equals Ω^0 by Hodge duality. All decompositions verified by direct projector computation (Algorithm 2) with multiplicities at machine precision. $\square$

Corollary 3.2. *Since D^2 is block-diagonal in sectors (Theorem 2.2) and the D_4 action preserves each sector (Proposition 2.3), the $\lambda = 4$ eigenspace of D^2 decomposes as*

$$V_{\lambda=4} = V_{\lambda=4}(\Omega^0) \oplus V_{\lambda=4}(\Omega^1) \oplus V_{\lambda=4}(\Omega^2), \quad (20)$$

with dimensions $14 + 28 + 14 = 56$. Each summand is D_4 -invariant.

4 The D_4 commutants

Definition 4.1 (Commutant). For a real vector space V with an orthogonal D_4 action,

$$\text{Comm}(D_4, V) = \{X \in \text{End}(V) : [X, R] = [X, S] = 0\}. \quad (21)$$

Theorem 4.2 (Commutant dimensions and structures). *By Schur's lemma,*

$$\text{Comm}(D_4, \Omega^0|_{\lambda=4}) = \Phi_3|_{A_1} \oplus \Phi_1|_{A_2} \oplus \Phi_2|_{B_1} \oplus \Phi_2|_{B_2} \oplus \Phi_3|_E, \quad \dim = 27, \quad (22)$$

$$\text{Comm}(D_4, \Omega^1|_{\lambda=4}) = \Phi_3|_{A_1} \oplus \Phi_3|_{A_2} \oplus \Phi_3|_{B_1} \oplus \Phi_3|_{B_2} \oplus \Phi_8|_E, \quad \dim = 100. \quad (23)$$

The commutant on the full 56-dimensional $\lambda = 4$ sector has dimension

$$9^2 + 5^2 + 7^2 + 7^2 + 14^2 = 400. \quad (24)$$

Proof. Schur's lemma: for an irrep ρ with multiplicity m_ρ , the commutant contributes $\mathfrak{gl}(m_\rho, \mathbb{R})$, with real dimension m_ρ^2 . For Ω^0 , multiplicities $(3, 1, 2, 2, 3)$ give $9 + 1 + 4 + 4 + 9 = 27$. For Ω^1 , $(3, 3, 3, 3, 8)$ give $9 + 9 + 9 + 9 + 64 = 100$. Numerical SVD of the constraint matrices gives exactly these dimensions with a clean spectral gap (Algorithm 3). $\square$

5 The gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \Phi_1^5$

Definition 5.1 ($\mathfrak{su}(3)_c$). Let $\{v_1, v_2, v_3\}$ be an orthonormal basis of the A_1 isotypic subspace of $\Omega^0|_{\lambda=4}$. Define the map $\phi : \mathfrak{gl}(3, \mathbb{R}) \rightarrow \text{End}(\mathbb{R}^{14})$ by

$$\phi(M) = \sum_{a,b=1}^3 v_a M_{ab} v_b^\top. \quad (25)$$

The eight generators of $\mathfrak{su}(3)_c$ are $T_c^a = \phi(\lambda^a)$ for the Gell-Mann matrices λ^a (traceless Hermitian 3×3):

$$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \dots \quad (26)$$

including the diagonal $\lambda^3 = \text{diag}(1, -1, 0)$ and $\lambda^8 = \text{diag}(1, 1, -2)/\sqrt{3}$.

Theorem 5.2 (Isomorphism $\mathfrak{sl}(3, \mathbb{R}) \cong \mathfrak{su}(3)_c$). *The map ϕ satisfies:*

- (i) ϕ is injective.
- (ii) $[\phi(M), R] = [\phi(M), S] = 0$ for all $M \in \mathfrak{gl}(3, \mathbb{R})$.
- (iii) $\phi([M, N]) = [\phi(M), \phi(N)]$.
- (iv) Restricted to traceless matrices, ϕ is a Lie algebra isomorphism onto $\mathfrak{su}(3)_c$.

Proof. (i) $\phi(M) = 0 \Rightarrow \sum_a v_a M_{ac} = 0$ for each c , hence $M = 0$. (ii) On the A_1 isotypic subspace, D_4 acts trivially; choose $\{v_a\}$ as a D_4 -invariant basis. (iii) Using $v_b^\top v_c = \delta_{bc}$:

$$[\phi(M), \phi(N)] = \sum_{a,d} v_a (MN - NM)_{ad} v_d^\top = \phi([M, N]). \quad (27)$$

(iv) Restricting to traceless matrices, the image is the traceless part of $\text{End}(V_{A_1})$, identified with $\mathfrak{su}(3)_c$ as a real Lie algebra. $\square$

Analogous constructions give $\mathfrak{su}(2)_L$ on the B_1 isotypic subspace with generators

$$T_L^i = \sum_{j,k=1}^2 w_j \sigma_{jk}^i w_k^\top, \quad i = 1, 2, 3, \quad (28)$$

for Pauli matrices σ^i and orthonormal basis $\{w_1, w_2\}$ of the B_1 isotypic.

The five Φ_1 factors are

$$Y_{A_1} = P_{A_1}, \quad Y_{A_2} = P_{A_2}, \quad Y_{B_1} = P_{B_1}, \quad Y_{B_2} = P_{B_2}, \quad Y_E = P_E. \quad (29)$$

Theorem 5.3 (Closure). *The 12 generators T_c^a (8), T_L^i (3), and a physical $\Phi_1)_Y$ (1) form a Lie subalgebra of $\text{Comm}(D_4, \Omega^0|_{\lambda=4})$. All closure residuals at machine precision: $\mathfrak{su}(3)_c$ at 4.05×10^{-16} , $\mathfrak{su}(2)_L$ at 4.02×10^{-16} , cross-commutators at $< 3 \times 10^{-16}$ (Algorithm 4).*

The same $\mathfrak{su}(3)_c$ and $\mathfrak{su}(2)_L$ act consistently on $\Omega^1|_{\lambda=4}$, with residuals $< 4 \times 10^{-16}$.

6 Nielsen-Ninomiya evasion

Theorem 6.1 (D and its spectrum). *Let $D = d + d^*$ on the discrete 2-torus $\mathbb{Z}_L \times \mathbb{Z}_L$, $L \geq 2$. Then:*

- (i) D is local (nearest-neighbour), Hermitian, and translation-invariant.
- (ii) $\dim(\ker D \cap \Omega^k) = b_k(T^2)$: for T^2 , $(b_0, b_1, b_2) = (1, 2, 1)$, total 4 zero modes.
- (iii) The nonzero spectrum of D is exactly $\pm$ -symmetric.

Proof. (i) D is the sum of nearest-neighbour hoppings, Hermitian since $d^* = d^\top$. (ii) $\ker D =$ harmonic forms on the torus. By Hodge theory, isomorphic to $H^\bullet(T^2; \mathbb{R})$, with dimensions $(b_0, b_1, b_2) = (1, 2, 1)$. Numerical verification at $L = 8$: zero modes split as $1 + 2 + 1$ (Algorithm 5). (iii) With $\gamma = (-1)^F$: $+1$ on $\Omega^0 \oplus \Omega^2$, -1 on Ω^1 . Then $\{\gamma, D\} = 0$, so $D\psi = \lambda\psi$ with $\lambda \neq 0$ implies $D(\gamma\psi) = -\lambda(\gamma\psi)$. Numerical: 126 positive and 126 negative eigenvalues. $\square$

Remark 6.2. The Nielsen-Ninomiya no-go assumes a bilinear lattice action on a *single* fermion species. The Hodge-Dirac violates this: it acts on the graded complex, with form degree as internal index. Zero modes are topological, not momentum-space doublers.

7 The vacuum zero state s

The zero state $s \in \mathbb{R}^{14}$ is the unique (up to scale) nullspace vector of the 14×19 vacuum-native measurement matrix M [Preschutti 2026], satisfying

$$M^\top s = 0 \tag{30}$$

to machine precision. Its components are listed in Table 1.

Mode	(n_1, n_2)	Orbit	s_i
ν	(0, 4)	$\mathcal{O}_0$	-0.105198
u	(1, 3)	$\mathcal{O}_1$	+0.937727
d	(1, 5)	$\mathcal{O}_1$	-0.192807
W	(2, 2)	$\mathcal{O}_2$	0
Z	(2, 6)	$\mathcal{O}_2$	-0.329142
s	(3, 1)	$\mathcal{O}_1$	-0.937727
μ	(3, 7)	$\mathcal{O}_1$	+0.658283
e	(4, 0)	$\mathcal{O}_0$	+0.105198
b	(5, 1)	$\mathcal{O}_1$	+0.192807
dark	(5, 7)	$\mathcal{O}_1$	-1.000000
H	(6, 2)	$\mathcal{O}_2$	+0.329142
t	(6, 6)	$\mathcal{O}_2$	0
c	(7, 3)	$\mathcal{O}_1$	-0.658283
τ	(7, 5)	$\mathcal{O}_1$	+1.000000

Table 1: The zero state s on the 14 modes at $\lambda = 4$. Normalized to $\max_i |s_i| = 1$.

Proposition 7.1 (Parity). $R_d s = -s$, where $R_d : (n_1, n_2) \mapsto (n_2, n_1)$ is the diagonal reflection.

Proof. Direct numerical verification: $\|R_d s + s\| = 0$ to machine precision. $\square$

Theorem 7.2 (Color singlet). $T_c^a s = 0$ for all $a = 1, \dots, 8$, to machine precision.

Proof. The A_1 irrep modes are even under R_d ; s is odd; hence $\langle A_1 \text{ mode}, s \rangle = 0$ identically. Therefore $P_{A_1} s = 0$, and since $\mathfrak{su}(3)_c$ acts only on the A_1 isotypic subspace, $T_c^a s = 0$. $\square$

Remark 7.3. Theorem 7.2 is a structural consequence of parity, not a numerical coincidence. This is the geometric origin of color conservation.

Proposition 7.4 (Weak doublet). $\|P_{B_1} s\| \approx 0.352 \neq 0$, so s is not a weak singlet: $\|T_L^i s\| \approx 0.249$ for each $\mathfrak{su}(2)_L$ generator.

8 Sector-triplet correspondence

Theorem 8.1 (Sector-triplet correspondence). *The 9 copies of the A_1 irrep in the 56-dimensional $\lambda = 4$ Hodge sector split as*

$$9A_1 = 3A_1(\Omega^0) \oplus 3A_1(\Omega^1) \oplus 3A_1(\Omega^2), \quad (31)$$

three copies in each Hodge sector. Each A_1 copy lies purely in one sector.

Proof. Follows from Corollary 3.2. Since D^2 is block diagonal in sectors and D_4 preserves each sector, the $\lambda = 4$ eigenspace is a direct sum $V_{\lambda=4}(\Omega^0) \oplus V_{\lambda=4}(\Omega^1) \oplus V_{\lambda=4}(\Omega^2)$, each D_4 -invariant. The A_1 isotypic of the direct sum is the direct sum of the A_1 isotypics. By Theorem 3.1, each summand contains 3 copies of A_1 . Numerical verification: purity to 6.7×10^{-16} (Algorithm 6). $\square$

Corollary 8.2 (Three color triplets per generation). *The 9 A_1 copies provide exactly three color triplets per generation, naturally labeled by Hodge sector.*

Corollary 8.3 (Arithmetic relation). *If the three color triplets carry hypercharges Y_0, Y_1, Y_2 ($\Omega^0, \Omega^1, \Omega^2$), the $\text{SU}(3)^2\text{U}(1)$ anomaly forces*

$$Y_0 + 2Y_1 + Y_2 = 0. \quad (32)$$

9 The orientation- $\mathbb{Z}_2$ theorem

The three Hodge sectors $\Omega^0, \Omega^1, \Omega^2$ are distinguished by three natural D_4 -invariant operators acting on the 56-dimensional $\lambda = 4$ sector:

Definition 9.1 (Sector operators).

$$\mathbb{I} = P_{\Omega^0} + P_{\Omega^1} + P_{\Omega^2} = \text{id}, \quad (33)$$

$$\mathcal{F} = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}, \quad (34)$$

$$\mathcal{O} = P_{\Omega^0} - P_{\Omega^2}. \quad (35)$$

$\mathcal{F}$ is the natural $\mathbb{Z}_2$ chirality grading of the Hodge complex ($\mathcal{F} = (-1)^F$). $\mathcal{O}$ is the orientation-reversing operator distinguishing Ω^0 from Ω^2 .

Definition 9.2 (Sector-coarse hypercharge). A *sector-coarse hypercharge* on the 56-dimensional $\lambda = 4$ Hodge sector is a D_4 -invariant functional of the form

$$Y = \sum_{\rho} (\alpha_{\rho} \mathbb{I} + \beta_{\rho} \mathcal{F} + \gamma_{\rho} \mathcal{O}) P_{\rho}, \quad (36)$$

where P_{ρ} is the projector onto the isotypic block of irrep ρ .

Theorem 9.3 (Orientation- Z_2 theorem). *Under the sector-coarse hypothesis and the requirement that Y reproduce the Standard Model per-generation content on the sector-triplets, the coefficients for the A_1 and B_1 blocks are uniquely determined:*

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = (0, -\frac{1}{6}, -\frac{1}{2}), \quad (37)$$

$$(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = (0, +\frac{1}{2}, +\frac{1}{2}). \quad (38)$$

These give the Standard Model hypercharges

$$Y(u_R^c) = -\frac{2}{3}, \quad Y(Q_L) = +\frac{1}{6}, \quad Y(d_R^c) = +\frac{1}{3}, \quad (39)$$

$$Y(e_R^c) = +1, \quad Y(L_L) = -\frac{1}{2}, \quad Y(\nu_R^c) = 0. \quad (40)$$

Proof. For the A_1 block, the sector assignment $\Omega^0 \rightarrow u_R^c$, $\Omega^1 \rightarrow Q_L$, $\Omega^2 \rightarrow d_R^c$ gives

$$\alpha_{A_1} + \beta_{A_1} + \gamma_{A_1} = -\frac{2}{3}, \quad (41)$$

$$\alpha_{A_1} - \beta_{A_1} = +\frac{1}{6}, \quad (42)$$

$$\alpha_{A_1} + \beta_{A_1} - \gamma_{A_1} = +\frac{1}{3}. \quad (43)$$

Solving gives $(\alpha, \beta, \gamma) = (0, -1/6, -1/2)$.

For the B_1 block, the sector assignment $\Omega^0 \rightarrow e_R^c$, $\Omega^1 \rightarrow L_L$, $\Omega^2 \rightarrow \nu_R^c$ gives

$$\alpha_{B_1} + \beta_{B_1} + \gamma_{B_1} = +1, \quad (44)$$

$$\alpha_{B_1} - \beta_{B_1} = -\frac{1}{2}, \quad (45)$$

$$\alpha_{B_1} + \beta_{B_1} - \gamma_{B_1} = 0. \quad (46)$$

Solving gives $(\alpha, \beta, \gamma) = (0, 1/2, 1/2)$. $\square$

Remark 9.4 (Structural features). The identity coefficients vanish ($\alpha_{A_1} = \alpha_{B_1} = 0$), meaning the mean hypercharge across the three sectors is zero. This is a structural feature of the SM content.

Remark 9.5 (Sector coarse hypothesis). The sector-coarse hypothesis is not derived from first principles. It states that the hypercharge functional acts uniformly on each isotypic block, with sector dependence captured only through the three natural operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$. This is the minimal assumption compatible with D_4 -invariance and the sector structure. Deriving the sector coarse hypothesis from deeper structure is an open problem.

10 Per generation matter content and anomaly cancellation

Theorem 10.1 (Standard Model per-generation content). *The Standard Model content per generation, in left-handed convention, is*

$$Q_L(3, 2, +\frac{1}{6}) \times 6, \quad u_R^c(3, 1, -\frac{2}{3}) \times 3, \quad d_R^c(3, 1, +\frac{1}{3}) \times 3, \quad (47)$$

$$L_L(1, 2, -\frac{1}{2}) \times 2, \quad e_R^c(1, 1, +1) \times 1, \quad \nu_R^c(1, 1, 0) \times 1, \quad (48)$$

for a total of 16 states per generation.

Theorem 10.2 (Anomaly cancellation). *The Standard Model content per generation satisfies all four anomaly cancellation conditions exactly:*

$$\text{Tr}(Y) = \text{Tr}(Y^3) = \text{SU}(2)^2\text{U}(1) = \text{SU}(3)^2\text{U}(1) = 0. \quad (49)$$

Proof. Direct computation using Dynkin indices $T(2) = T(3) = 1/2$:

$$\text{Tr}(Y) = 6 \cdot \frac{1}{6} + 3 \cdot \left(-\frac{2}{3}\right) + 3 \cdot \frac{1}{3} + 2 \cdot \left(-\frac{1}{2}\right) + 1 \cdot 1 + 0 = 0, \quad (50)$$

$$\text{Tr}(Y^3) = 6 \cdot \left(\frac{1}{6}\right)^3 + 3 \cdot \left(-\frac{2}{3}\right)^3 + 3 \cdot \left(\frac{1}{3}\right)^3 + 2 \cdot \left(-\frac{1}{2}\right)^3 + 1 = 0, \quad (51)$$

$$\text{SU}(2)^2\text{U}(1) = \frac{1}{2} \cdot \left[3 \cdot \frac{1}{6} \cdot 2 + 1 \cdot \left(-\frac{1}{2}\right) \cdot 2\right] \cdot 2 = 0, \quad (52)$$

$$\text{SU}(3)^2\text{U}(1) = \frac{1}{2} \cdot \left[\frac{1}{6} \cdot 2 + \left(-\frac{2}{3}\right) \cdot 1 + \frac{1}{3} \cdot 1\right] \cdot 2 = 0. \quad (53)$$

□

Corollary 10.3 (Hypercharge arithmetic progression). *For the three color triplets, the SM hypercharges satisfy*

$$Y_{u_R^c} + 2Y_{Q_L} + Y_{d_R^c} = -\frac{2}{3} + \frac{2}{6} + \frac{1}{3} = 0, \quad (54)$$

consistent with the sector-triplet correspondence.

11 Non-equivariance of the Standard Model hypercharge

Theorem 11.1 (Non-equivariance). *Any D_4 -equivariant hypercharge Y on any D_4 -invariant subspace of the Hodge complex is constant on each isotypic block. If the content contains exactly one color triplet (A_1) and one weak doublet (B_1), the anomaly conditions $\text{SU}(3)^2\text{U}(1) = \text{SU}(2)^2\text{U}(1) = 0$ force $Y|_{A_1} = Y|_{B_1} = 0$.*

Proof. D_4 -equivariance and Schur's lemma give $Y|_{V_\rho} = c_\rho \cdot \mathbb{I}_{V_\rho}$. For one color triplet and one weak doublet, the anomaly coefficients are $\text{SU}(3)^2\text{U}(1) = 3c_{A_1}$ and $\text{SU}(2)^2\text{U}(1) = 2c_{B_1}$, forcing both to zero. □

Corollary 11.2 (SM hypercharge is non-equivariant). *The Standard Model hypercharge assigns distinct values to Q_L ($+1/6$), u_R ($+2/3$), and d_R ($-1/3$), all in the same color isotypic block. Therefore no D_4 -equivariant hypercharge assignment reproduces the SM pattern.*

12 What is derived and what remains open

12.1 Established results

- (i) Hodge–Dirac structure: $D = d + d^*$ with $D^2 = \Delta$ exactly.
- (ii) $\lambda = 4$ content: $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$.
- (iii) Commutants: $\dim 27$, $\dim 100$, $\dim 400$.
- (iv) Gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{P}_1^5$ as an exact Lie subalgebra.
- (v) Nielsen–Ninomiya evasion.
- (vi) Vacuum structure: color singlet (parity-forced), weak doublet.
- (vii) Sector-triplet correspondence: 9 A_1 copies split 3/3/3.
- (viii) Orientation- Z_2 theorem: SM hypercharges uniquely determined under sector-coarse hypothesis.
- (ix) Per-generation SM content with all four anomaly conditions vanishing exactly.
- (x) Non-equivariance theorem.

12.2 Open problems

Open Problem 12.1 (Sector-coarse hypothesis). Derive the sector-coarse hypothesis (that the hypercharge functional is a linear combination of $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$) from deeper structure, such as the FDLM/WIN processor’s action on the 56-dimensional content.

Open Problem 12.2 (Three-generation structure). Establish that the three spin structures of the torus generate three generations via the $\mathbb{Z}_3$ permutation subgroup of S_4 or A_4 , and verify consistency with the gauge and matter structure derived here.

Open Problem 12.3 (Yukawa couplings and mass generation). Derive the Yukawa coupling structure from the geometric data on the 56-dimensional Hodge sector, and verify against observed masses.

Open Problem 12.4 (V_{td} anomaly). The CKM prediction $V_{td} = 0.00613$ differs from the unitarity-constrained value 0.0086. Whether the A_4 extension resolves this is open.

Open Problem 12.5 (Strong CP). Whether the framework’s $\mathbb{Z}_2$ structure constrains the QCD θ parameter to zero is open.

Open Problem 12.6 (Full two-loop anomaly). Verify that anomaly cancellation on the framework content holds at two loops.

13 Conclusion

We have derived the Standard Model gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$, the per-generation matter content, and the Standard Model hypercharges from the Hodge complex of differential forms on the 8×8 torus. The derivation yields the gauge algebra as an exact Lie subalgebra of the D_4 commutant, the vacuum structure as the zero state s (color singlet, weak doublet), the sector-triplet correspondence as an analytic consequence of the sector-block structure of D^2 , and the SM hypercharge assignment as the unique orientation- $\mathbb{Z}_2$ decomposition under the sector-coarse hypothesis. All four anomaly cancellation conditions vanish exactly.

A structural theorem establishes that the Standard Model hypercharge is not D_4 -equivariant, and the resolution comes through the sector-resolved content of the full Hodge complex. The framework provides not just the gauge algebra but the specific sector structure through which the SM content emerges.

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A Numerical algorithms

All numerical claims in this paper are reproducible with the following algorithms. Each algorithm is stated in pseudocode; complete Python implementations are available in the companion repository.

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Algorithm 1 Build Hodge–Dirac and D_4 action**Require:** lattice size $L = 8$, $N = L^2$, $N_{\text{edges}} = 2N$

```

1: Initialize  $d_0 \in \mathbb{R}^{N_{\text{edges}} \times N}$ ,  $d_1 \in \mathbb{R}^{N \times N_{\text{edges}}}$  as zero
2: for  $x_1, x_2 \in \{0, \dots, L-1\}$  do
3:    $i \leftarrow \text{site}(x_1, x_2)$ 
4:   for  $\mu \in \{0, 1\}$  do
5:      $j \leftarrow \text{edge}(x_1, x_2, \mu)$ 
6:      $d_0[j, i] -= 1$ 
7:      $d_0[j, \text{site}(x_1 + [\mu = 0], x_2 + [\mu = 1])] += 1$ 
8:   end for
9: end for
10: ▷ Similarly for  $d_1$  with signs
11:  $D \leftarrow \begin{pmatrix} 0 & d_0^\top & 0 \\ d_0 & 0 & d_1^\top \\ 0 & d_1 & 0 \end{pmatrix}$ 
12: ▷ Build  $R, S$  as permutation matrices with vector twist on  $\Omega^1$ 
13: return  $D, R, S$ 

```

Algorithm 2 D_4 irrep projector**Require:** D_4 generators R, S on vector space V of dimension d **Require:** irrep name $\rho \in \{A_1, A_2, B_1, B_2, E\}$

```

1: Compute  $R^2, R^3, RS, R^2S, R^3S$ 
2: Set elements  $\leftarrow [I, R, R^2, R^3, S, RS, R^2S, R^3S]$ 
3: Set chars[ $\rho$ ] from  $D_4$  character table
4:  $P \leftarrow \sum_k \text{chars}[\rho][k] \cdot \text{elements}[k]$ 
5: return  $d_\rho \cdot P/8$ 

```

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Algorithm 3 Commutant of D_4 via Schur

Require: D_4 generators R, S on V of dimension d

- 1: Build constraint matrix $A = \begin{pmatrix} I \otimes R - R^\top \otimes I \\ I \otimes S - S^\top \otimes I \end{pmatrix}$
 - 2: Compute nullspace of A via SVD: $\text{null}(A) = \{x : Ax = 0\}$
 - 3: $\text{dim Comm} \leftarrow \text{shape}(\text{null}(A))[1]$
 - 4: **return** dim Comm , basis of $\text{null}(A)$
-

Algorithm 4 Gauge algebra closure check

Require: Generators $\{T^a\}_{a=1}^n$ (Hermitian)

- 1: Gram–Schmidt orthonormalize $\{T^a\}$ under trace inner product
 - 2: $\text{max_residual} \leftarrow 0$
 - 3: **for** each pair (T^a, T^b) **do**
 - 4: $[T^a, T^b] \leftarrow T^a T^b - T^b T^a$
 - 5: Project: $\Pi \leftarrow \sum_c \text{Tr}(T^c(-i[T^a, T^b]))T^c$
 - 6: $\text{residual} \leftarrow \|[T^a, T^b] - i\Pi\|$
 - 7: Update max_residual
 - 8: **end for**
 - 9: **return** max_residual
-

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Algorithm 5 Nielsen–Ninomiya zero-mode check**Require:** Hodge–Dirac D on 256-dim complex

- 1: Compute eigenvalues of D : λ_i, v_i
- 2: Zero modes: $\{v_i : |\lambda_i| < 10^{-10}\}$
- 3: **for** each zero mode v **do**
- 4: Compute sector projections $p_0 = \|v[0 : 64]\|^2$, $p_1 = \|v[64 : 192]\|^2$, $p_2 = \|v[192 : 256]\|^2$
- 5: Assign to sector with $\max(p_0, p_1, p_2)$
- 6: **end for**
- 7: Verify: sector counts $(1, 2, 1)$ for $(\Omega^0, \Omega^1, \Omega^2)$
- 8: Verify: number of positive eigenvalues = number of negative
- 9: **return** sector counts, $\pm$ symmetry

Algorithm 6 Sector-triplet correspondence**Require:** A_1 isotypic basis $\{v_k\}_{k=1}^9$ in 56-dim $\lambda = 4$ subspace**Require:** Embedding $V_{\lambda=4} \subset \mathbb{R}^{256}$

- 1: **for** each k **do**
- 2: Compute $v_k^{256} \leftarrow V_{\lambda=4} v_k$
- 3: Compute sector projections p_0, p_1, p_2
- 4: Assign to $\arg \max(p_0, p_1, p_2)$
- 5: Verify cross-sector leakage $< 10^{-14}$
- 6: **end for**
- 7: **return** sector assignment $(3, 3, 3)$

Algorithm 7 Orientation- Z_2 hypercharge decomposition**Require:** Sector projectors $P_{\Omega^0}, P_{\Omega^1}, P_{\Omega^2}$

- 1: $\mathcal{F} \leftarrow P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}$
- 2: $\mathcal{O} \leftarrow P_{\Omega^0} - P_{\Omega^2}$
- 3: Define sector values $\mathcal{F}_{\Omega^0} = 1$, $\mathcal{F}_{\Omega^1} = -1$, $\mathcal{F}_{\Omega^2} = 1$; $\mathcal{O}_{\Omega^0} = 1$, $\mathcal{O}_{\Omega^1} = 0$, $\mathcal{O}_{\Omega^2} = -1$
- 4: Set up linear system for (α, β, γ) given SM hypercharges $Y_{\Omega^0}, Y_{\Omega^1}, Y_{\Omega^2}$:
- 5: $\alpha + \beta \mathcal{F}_{\Omega^0} + \gamma \mathcal{O}_{\Omega^0} = Y_{\Omega^0}$
- 6: $\alpha + \beta \mathcal{F}_{\Omega^1} + \gamma \mathcal{O}_{\Omega^1} = Y_{\Omega^1}$
- 7: $\alpha + \beta \mathcal{F}_{\Omega^2} + \gamma \mathcal{O}_{\Omega^2} = Y_{\Omega^2}$
- 8: Solve linear system
- 9: **return** (α, β, γ)

Algorithm 8 Anomaly cancellation check**Require:** Content table: list of (d_3, d_2, Y) triples

- 1: $\text{Tr}Y \leftarrow \sum_i d_3^{(i)} d_2^{(i)} Y_i$
- 2: $\text{Tr}Y^3 \leftarrow \sum_i d_3^{(i)} d_2^{(i)} Y_i^3$
- 3: $\text{SU}(2)^2\text{U}(1) \leftarrow \sum_{i:d_2=2} T(2) \cdot d_3^{(i)} \cdot Y_i$
- 4: $\text{SU}(3)^2\text{U}(1) \leftarrow \sum_{i:d_3=3} T(3) \cdot d_2^{(i)} \cdot Y_i$
- 5: **return** $(\text{Tr}Y, \text{Tr}Y^3, \text{SU}(2)^2\text{U}(1), \text{SU}(3)^2\text{U}(1))$